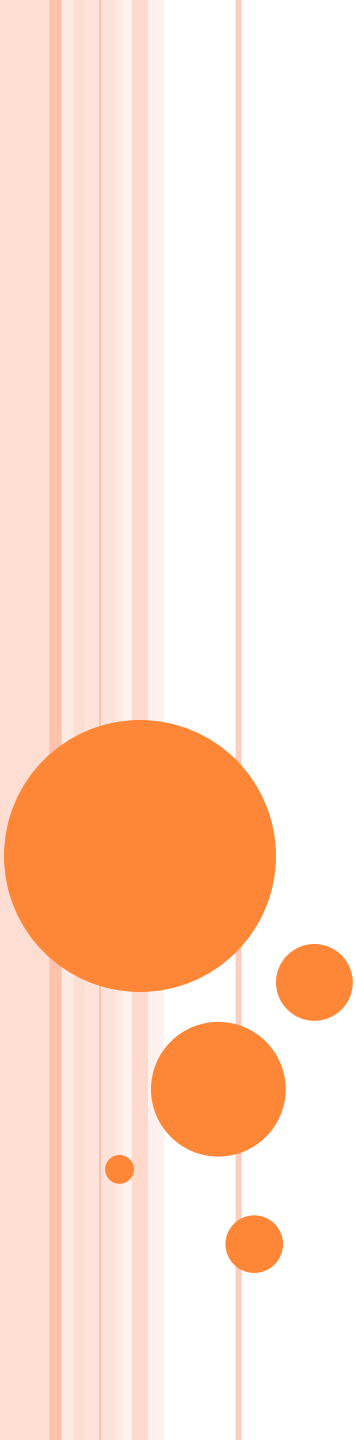




Exoplanet Hunter: ML-Based Transit Detection Using Real NASA Kepler/TESS Data

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Introduction

- ❑ **What is an Exoplanet?**
- ❑ An **exoplanet** is a planet located outside our Solar System.
- ❑ One way of detecting an exoplanet is through the **transit method**.
- ❑ When a planet passes between its host star and the observer:
- ❑ Star ★ ← Planet ● ← Telescope
- ❑ the observed brightness of the star decreases slightly.
- ❑ This creates a characteristic **dip in the light curve**.



Problem Statement

Astronomical missions generate a huge amount of stellar brightness data.

Manually examining these light curves is:

- Time-consuming
- Difficult for large datasets
- Sensitive to noise
- Difficult to automate

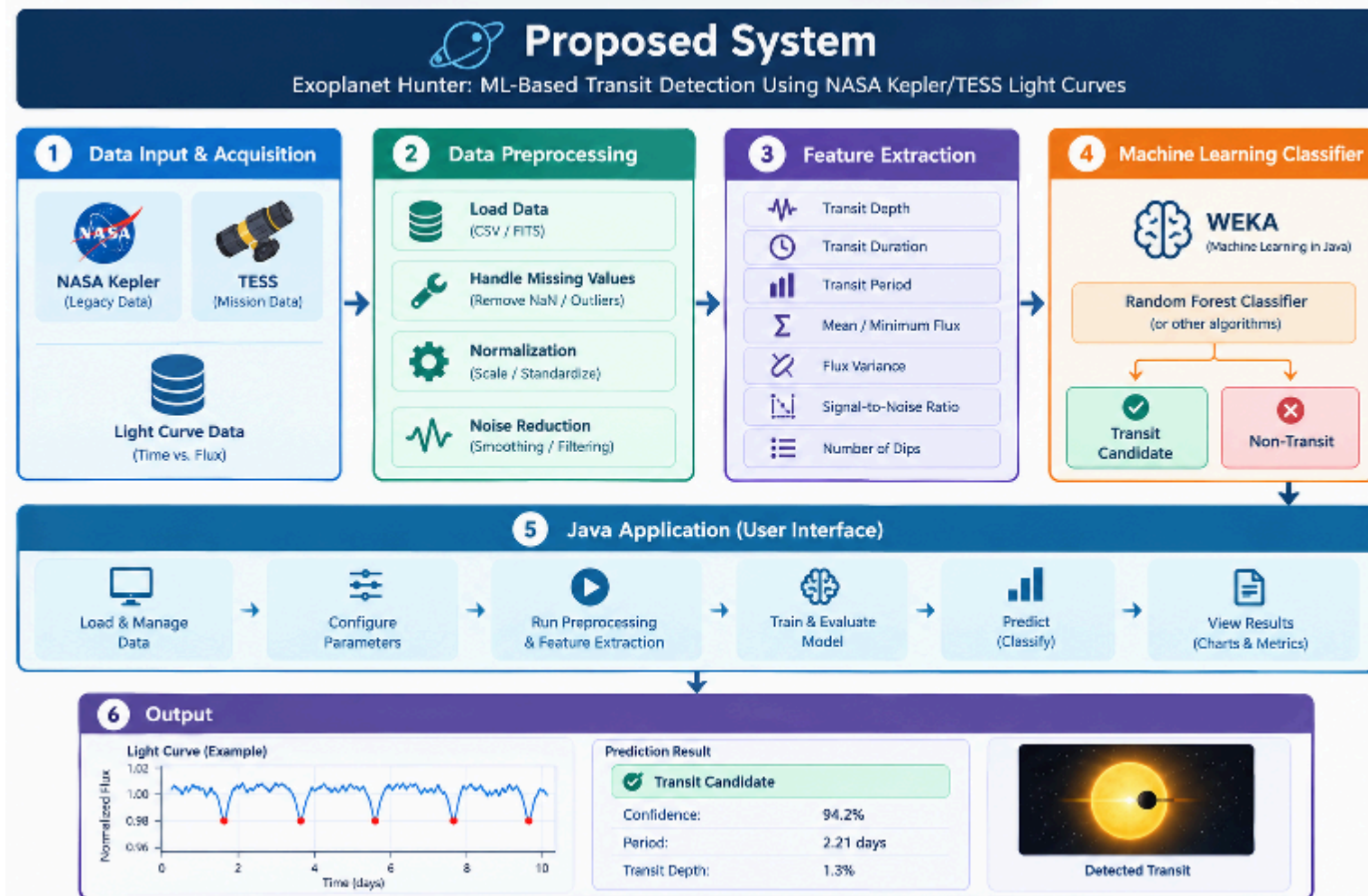


Objective

- Obtain real NASA Kepler/TESS light-curve data.
- Preprocess and normalize the data.
- Extract useful transit features.
- Apply a machine-learning classifier.
- Display whether a candidate is likely to contain a transit.



Proposed System



Dataset

NASA Kepler / TESS Data

- The project will use real astronomical observations from NASA archives.
- The TESS archive provides light-curve files containing measured flux over observation times, and these products can be accessed through MAST.
- Kepler's archive also provides processed light-curve files and associated target data.

Data contains

- Time
- Flux
- Flux Error
- Target ID
- Sector / Quarter



ML Approach

- Input: stellar light curve
- Normalize flux
- Detect/segment possible transit regions
- Extract features such as:
 - Transit depth
 - Duration
 - Periodicity
 - Number of dips
- Classifier:
 - Random Forest / SVM
 - or a simple Neural Network



Java Technology Stack

- Java
- JavaFX — GUI and graphs
- Apache Commons Math — numerical processing
- Tribuo / Weka — machine learning
- FITS data parser/library
- CSV for intermediate datasets



Prototype GUI

- You can show a mockup containing:
- **Upload Light Curve** → **Analyze** → graph of brightness vs time →
Prediction: TRANSIT CANDIDATE
Confidence: 94%



Expected Output

- Automatically identify potential transit signals.
- Visualize stellar brightness variations.
- Reduce manual inspection.
- Provide an educational demonstration of ML applied to astronomy.



Future Scope

- Support both Kepler and TESS.
- Deep-learning models.
- Larger datasets.
- Automatic candidate ranking.
- Integration with NASA Exoplanet Archive.

